

CONFIDENCE INTERVALS

master recipe • also used in Units 4 and 5

ZONE A · THE FORMULAS

statistic ±

(critical value)(standard error)

margin of error

ME = z^* · SE (proportions)

ME = t^* · SE (means)

Use the procedure’s df to choose t^* .

width $\propto \frac{1}{\sqrt{n}}$ $\Rightarrow$ halve the margin $\Rightarrow$ quadruple n

THE FOUR-STEP FRAME

1	STATE	Name the parameter in context.
2	PLAN	Name the interval; check conditions.
3	DO	Formula, substitutions, and interval.
4	CONCLUDE	Interpret the interval in context.

COMMON z^* VALUES

confidence	z^*
90%	1.645
95%	1.960
99%	2.576

Get t^* from the table or calculator using **df**.

ZONE B · CONDITIONS — ALWAYS IN THIS ORDER

1 · RANDOM

Random sample or
randomized experiment.

2 · 10%

Sampling without replacement:
 $n < 0.10N$.

3 · NORMAL / LARGE COUNTS

CI: $n\hat{p} \geq 10, n(1 - \hat{p}) \geq 10$
Test: $np_0 \geq 10, n(1 - p_0) \geq 10$ (see P11)
means: normal population, $n \geq 30$, or no strong
skew/outliers

ZONE C · SAY IT IN WORDS

“We are _____% confident that the interval from _____ to _____ captures the true _____.”

Confidence level: If we took many samples and built an interval each time,
about _____% of those intervals would capture the true _____.

ZONE D · WATCH OUT

THE INTERVAL ESTIMATES THE PARAMETER. THE LEVEL DESCRIBES THE METHOD.

Never say the fixed parameter has a 95% chance of being inside the interval. After the interval is calculated,
it either captures the parameter or it does not.